

Weeksolved

Computer program source code

Author and copyright claimant: Alberto Pelayo

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IDENTIFYING MATERIAL

This work contains trade secret material. In accordance with the deposit options for computer programs containing trade secrets, this deposit consists of the first ten (10) pages and the last ten (10) pages of source code, submitted without any portion blocked out.

The complete source code totals approximately 278 printed pages.
The page bearing the copyright notice is included as page 1.

```

<!-- Weeksolved. Copyright (c) 2026 Alberto Pelayo. All rights reserved. -->
<title>Weeksolved – Staff Scheduling Software with Break Compliance</title>
<meta name="description" content="Weeksolved builds your staff schedule from your own rules: required coverage, role ratios, availability, hour caps and state meal and rest break law. Weekly and daily rotas, station sheets, holiday polls, Excel and PDF export.">
<meta name="keywords" content="employee scheduling software, staff scheduling, shift scheduler, work schedule maker, shift planner, staff rota software, employee rota, break compliance, meal break law, rest break scheduling, California meal break, labor law scheduling, coverage planning, shift rotation, seniority scheduling, healthcare staff scheduling, pharmacy staff scheduling, clinic scheduling software, small business scheduling">
<meta name="author" content="Weeksolved">
<meta name="robots" content="index,follow">
<meta name="theme-color" content="#1c6a66">
<meta property="og:type" content="website">
<meta property="og:site_name" content="Weeksolved">
<link rel="canonical" href="https://weeksolved.com/app.html">
<meta property="og:url" content="https://weeksolved.com/app.html">
<meta property="og:title" content="Weeksolved – Staff Scheduling Software with Break Compliance">
<meta property="og:description" content="Give it your coverage rules once. Weeksolved drafts a compliant weekly schedule, places every meal and rest break inside the legal window, and tells you exactly what blocked any shift it could not fill.">
<meta name="twitter:card" content="summary_large_image">
<meta name="twitter:title" content="Weeksolved – Staff Scheduling Software with Break Compliance">
<meta name="twitter:description" content="Staff scheduling that follows your rules and your state's break law. Weekly and daily rotas, station sheets, fair rotation, Excel and PDF export.">
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "SoftwareApplication",
  "name": "Weeksolved",
  "url": "https://weeksolved.com/",
  "alternateName": "Weeksolved Staff Scheduling",
  "applicationCategory": "BusinessApplication",
  "applicationSubCategory": "Employee Scheduling Software",
  "operatingSystem": "Any modern web browser",
  "description": "Weeksolved turns a manager's coverage rules into a compliant staff schedule. It handles required coverage by role and time of day, role ratios, staff groups and hour caps, individual availability and requests, seniority or lottery rotation for the sought-after shifts, and meal and rest breaks under the labor law of the state you pick. It produces a weekly grid, an hour-by-hour daily station sheet, Excel and PDF exports, and a ranked-choice holiday poll.",
  "featureList": [
    "Automatic weekly schedule generation from your own rules",
    "State meal and rest break law applied to every shift",
    "Required coverage minimums, role ratios and pairing rules",
    "Staff groups with their own hour, day and shift limits",
    "Employee availability, preferences and time off",
    "Requests inbox with a plain-English outcome for each one",
    "Fair rotation of the sought-after shifts by seniority, fairness or lottery",
    "Daily station sheet with meal and rest break times",
    "Coverage-gap solution naming who could cover and at what cost",
    "Ranked-choice holiday poll",
    "Excel (CSV) and PDF export"
  ],
  "offers": {
    "@type": "Offer",
    "price": "0",
    "priceCurrency": "USD"
  },
  "audience": {
    "@type": "BusinessAudience",
    "audienceType": "Managers and schedulers in pharmacies, clinics, laboratories, veterinary practices, retail, hospitality, security and warehousing"
  }
}
</script>
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "FAQPage",
  "mainEntity": [
    {
      "@type": "Question",
      "name": "Does Weeksolved follow state meal and rest break law?",

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    "acceptedAnswer":{"@type":"Answer","text":"Yes. You pick your jurisdiction and Weeksolved applies its meal
      and rest break rules to every shift it schedules \u2014 how long a shift has to be before a meal is
      owed, the latest legal moment it may begin, how many paid rest breaks are owed for the hours worked
      , and whether a waiver is permitted. It flags a late or missing meal period before you post the sche
      dule."}},
    {"@type":"Question","name":"Will it invent shifts I never asked for?",
      "acceptedAnswer":{"@type":"Answer","text":"No. Weeksolved only ever schedules shifts you have defined. If
        your coverage cannot be met from that pool it leaves the slot open and tells you exactly which limit
        blocked each candidate, rather than quietly bending a rule to fill it."}},
    {"@type":"Question","name":"Can it handle part-time, per diem and contracted-hours staff?",
      "acceptedAnswer":{"@type":"Answer","text":"Yes. Each person can work to a range or to an exact contracted
        figure, with their own day and hour limits. Per diem staff are kept out of the draft and offered whe
        n a shift cannot otherwise be covered. Staff groups can tighten limits across a whole category at on
        ce."}},
    {"@type":"Question","name":"How does it decide who gets the good shifts?",
      "acceptedAnswer":{"@type":"Answer","text":"You decide. Weeksolved can weight the sought-after shifts by se
        niority, spread them evenly, run an auditable lottery, or ignore standing altogether. Whichever you
        choose, it keeps a ledger so the same people do not draw the short straw week after week."}}
  ]}
</script>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link rel="stylesheet" href="https://fonts.googleapis.com/css2?family=Archivo:wght@500;600;700&family=Public
  +Sans:wght@400;500;600&family=IBM+Plex+Mono:wght@400;500;600&display=swap">
<style>
:root{
  --ground:#eadeb; --surface:#ffffff; --surface2:#f4f6f5; --sunken:#e2e6e4;
  --ink:#18211f; --ink2:#3c4a47; --muted:#697773; --faint:#667470;
  --line:#d2d8d6; --line2:#bfc7c4; --line-strong:#7d8b87;
  --accent:#1c6a66; --accent-soft:#dcebe9; --accent-ink:#0f4a47; --on-accent:#ffffff;
  --ok:#2c7a4b; --ok-soft:#dcefe2;
  --warn:#8f5d10; --warn-soft:#f6ead4; --on-warn:#ffffff;
  --crit:#ab382c; --crit-soft:#f7e0dc;
  --shadow:0 1px 2px rgba(24,33,31,.06),0 6px 18px rgba(24,33,31,.05);
  --r:7px;
}
@media (prefers-color-scheme: dark){
:root:not([data-theme="light"]){
  --ground:#101514; --surface:#191f1e; --surface2:#212927; --sunken:#0b100f;
  --ink:#e7ecea; --ink2:#c2ccc9; --muted:#8d9b97; --faint:#6d7b77; --line-strong:#778682; --line-strong:#7
    78682;
  --line:#2c3634; --line2:#3c4846;
  --accent:#4cb3a9; --accent-soft:#173432; --accent-ink:#8fd8d0; --on-accent:#08201f;
  --ok:#5cb87e; --ok-soft:#16301f;
  --warn:#d9a441; --warn-soft:#33260f; --on-warn:#241905; --on-warn:#241905;
  --crit:#e2796a; --crit-soft:#3a1c17;
  --shadow:0 1px 2px rgba(0,0,0,.4),0 8px 24px rgba(0,0,0,.32);
}
}
:root[data-theme="dark"]{
  --ground:#101514; --surface:#191f1e; --surface2:#212927; --sunken:#0b100f;
  --ink:#e7ecea; --ink2:#c2ccc9; --muted:#8d9b97; --faint:#6d7b77;

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--line:#2c3634; --line2:#3c4846;
--accent:#4cb3a9; --accent-soft:#173432; --accent-ink:#8fd8d0; --on-accent:#08201f;
--ok:#5cb87e; --ok-soft:#16301f;
--warn:#d9a441; --warn-soft:#33260f; --on-warn:#241905;
--crit:#e2796a; --crit-soft:#3a1c17;
--shadow:0 1px 2px rgba(0,0,0,.4),0 8px 24px rgba(0,0,0,.32);
}

*{box-sizing:border-box}
body{
  margin:0; background:var(--ground); color:var(--ink);
  font-family:"Public Sans",-apple-system,BlinkMacSystemFont,"Segoe UI",Roboto,sans-serif;
  font-size:14px; line-height:1.5; -webkit-font-smoothing:antialiased;
}
h1,h2,h3,h4{font-family:"Archivo","Public Sans",sans-serif;margin:0;text-wrap:balance;letter-spacing:-.012em
}
.mono{font-family:"IBM Plex Mono",ui-monospace,SFMono-Regular,Menlo,monospace;font-variant-numeric:tabular-n
ums}
button,input,select,textarea{font:inherit;color:inherit}
:focus-visible{outline:2px solid var(--accent);outline-offset:2px;border-radius:4px}
@media (prefers-reduced-motion:reduce){*{animation-duration:.001ms!important;transition-duration:.001ms!impo
rtant}}

/* ----- shell ----- */
.app{display:grid;grid-template-columns:216px 1fr;min-height:100vh}
.rail{
  background:var(--surface);border-right:1px solid var(--line);
  display:flex;flex-direction:column;position:sticky;top:0;height:100vh;overflow-y:auto;
}
.brand{padding:18px 16px 14px;border-bottom:1px solid var(--line);display:flex;gap:10px;align-items:center}
.brand .mark{
  width:30px;height:30px;border-radius:6px;background:var(--accent);color:var(--on-accent);
  display:grid;place-items:center;font-family:"Archivo","Public Sans",sans-serif;font-weight:700;font-size:1
  5px;flex:none;letter-spacing:-.02em;
}
.brand h1{font-size:15px;font-weight:700;line-height:1.15}
.brand .sub{font-size:10.5px;color:var(--muted);text-transform:uppercase;letter-spacing:.09em;font-weight:50
0}
.nav{padding:10px 8px;display:flex;flex-direction:column;gap:1px;flex:1}
.nav .grp{font-size:10px;text-transform:uppercase;letter-spacing:.1em;color:var(--faint);font-weight:600;pad
ding:14px 8px 5px}
.nav button{
  display:flex;align-items:center;gap:9px;width:100%;text-align:left;background:none;border:0;
  padding:8px 9px;border-radius:6px;cursor:pointer;color:var(--ink2);font-size:13.5px;font-weight:500;
}
.nav button:hover{background:var(--surface2);color:var(--ink)}
.nav button[aria-current="true"]{background:var(--accent-soft);color:var(--accent-ink);font-weight:600}
.nav .n{margin-left:auto;font-size:11px;color:var(--muted);font-family:"IBM Plex Mono",monospace}
.nav button[aria-current="true"] .n{color:var(--accent-ink)}
.railfoot{padding:10px;border-top:1px solid var(--line);display:flex;flex-wrap:wrap;gap:6px;align-items:cent
er}
.dot{width:7px;height:7px;border-radius:50%;background:var(--faint);flex:none}

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.dot.on{background:var(--ok)}
.railfoot small{color:var(--muted);font-size:11px}

.main{min-width:0;display:flex;flex-direction:column}
.topbar{
  position:sticky;top:0;z-index:20;background:var(--surface);border-bottom:1px solid var(--line);
  padding:14px 22px;display:flex;align-items:center;gap:14px;flex-wrap:wrap;
}
.topbar h2{font-size:18px;font-weight:600}
.topbar .desc{color:var(--muted);font-size:12.5px;flex:1;min-width:160px}
.pane{padding:22px;max-width:1240px;width:100%}

/* ----- primitives ----- */
.card{background:var(--surface);border:1px solid var(--line);border-radius:var(--r);box-shadow:var(--shadow)}
}
.card>header{padding:13px 16px;border-bottom:1px solid var(--line);display:flex;align-items:center;gap:10px;
  flex-wrap:wrap}
.card>header h3{font-size:13.5px;font-weight:600}
.card>header .hint{font-size:12px;color:var(--muted);flex:1;min-width:120px}
.card .body{padding:16px}
.stack{display:flex;flex-direction:column;gap:16px}
.row{display:flex;gap:10px;align-items:center;flex-wrap:wrap}
.spread{display:flex;gap:10px;align-items:center;justify-content:space-between;flex-wrap:wrap}
.grid2{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:12px}
.grid3{display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px}
/* a two-up layout asked for by name still has to fold on a phone */
@media (max-width:700px){.grid2{grid-template-columns:minmax(0,1fr)!important}}
/* let a wrapping row inside a spread actually wrap instead of pushing wide */
.spread>*.row>.row{min-width:0}
.field{display:flex;flex-direction:column;gap:5px;min-width:0}
.field>label,.lab{font-size:11px;text-transform:uppercase;letter-spacing:.07em;color:var(--muted);font-weight:600}
input[type=text],input[type=number],input[type=time],input[type=date],input[type=email],input[type=tel],select,textarea{
  background:var(--surface);border:1px solid var(--line-strong);border-radius:5px;padding:7px 9px;width:100%;
  transition:border-color .12s,box-shadow .12s;
}
input:focus,select:focus,textarea:focus{outline:none;border-color:var(--accent);box-shadow:0 0 3px var(--accent-soft)}
textarea{resize:vertical;min-height:60px;font-size:13px}
input[type=time],input[type=date],input[type=number]{font-family:"IBM Plex Mono",monospace;font-variant-numeric:tabular-nums}

.btn{
  display:inline-flex;align-items:center;gap:6px;border:1px solid var(--line2);background:var(--surface);
  padding:7px 12px;border-radius:6px;cursor:pointer;font-weight:500;font-size:13px;white-space:nowrap;
  transition:background .12s,border-color .12s,transform .06s;
}
.btn:hover{background:var(--surface2);border-color:var(--muted)}
.btn:active{transform:translateY(1px)}
.btn.primary{background:var(--accent);color:var(--on-accent);border-color:var(--accent);font-weight:600}
.btn.primary:hover{filter:brightness(1.08)}

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.btn.ghost{border-color:transparent;background:none;color:var(--muted)}
.btn.ghost:hover{background:var(--surface2);color:var(--ink)}
.btn.danger{color:var(--crit);border-color:var(--line2)}
.btn.danger:hover{background:var(--crit-soft);border-color:var(--crit)}
.btn.sm{padding:4px 8px;font-size:12px;border-radius:5px}
.btn.disabled{opacity:.45;cursor:not-allowed}
.chip{
  display:inline-flex;align-items:center;gap:5px;padding:2px 8px;border-radius:99px;font-size:11.5px;font-weight:600;
  background:var(--surface2);color:var(--ink2);border:1px solid var(--line);white-space:nowrap;
}
.chip.ok{background:var(--ok-soft);color:var(--ok);border-color:transparent}
.chip.warn{background:var(--warn-soft);color:var(--warn);border-color:transparent}
.chip.crit{background:var(--crit-soft);color:var(--crit);border-color:transparent}
.chip.acc{background:var(--accent-soft);color:var(--accent-ink);border-color:transparent}
.tag{display:inline-block;padding:1px 6px;border-radius:4px;font-size:10.5px;font-weight:600;letter-spacing:.03em}
table{border-collapse:collapse;width:100%;font-size:13px}
th{text-align:left;font-size:10.5px;text-transform:uppercase;letter-spacing:.07em;color:var(--muted);font-weight:600;padding:8px 10px;border-bottom:1px solid var(--line)}
td{padding:8px 10px;border-bottom:1px solid var(--line)}
tr:last-child td{border-bottom:0}
.scroll{overflow-x:auto;overscroll-behavior-x:contain}
.empty{padding:30px 18px;text-align:center;color:var(--muted);font-size:13px}
.empty strong{display:block;color:var(--ink);font-weight:600;margin-bottom:4px;font-family:"Archivo","Public Sans",sans-serif}
.help{font-size:12px;color:var(--muted);line-height:1.5}
.sep{height:1px;background:var(--line);margin:4px 0}
.switch{display:inline-flex;align-items:center;gap:8px;cursor:pointer;font-size:13px;user-select:none}
.switch input{appearance:none;width:34px;height:19px;border-radius:99px;background:var(--sunken);border:1px solid var(--line2);position:relative;cursor:pointer;transition:background .15s;flex:none;margin:0}
.switch input:checked{background:var(--accent);border-color:var(--accent)}
.switch input::after{content:"";position:absolute;top:1px;left:1px;width:15px;height:15px;border-radius:50%;background:var(--surface);transition:transform .15s;box-shadow:0 1px 2px rgba(0,0,0,.2)}
.switch input:checked::after{transform:translateX(15px)}
.seg{display:inline-flex;background:var(--sunken);border-radius:6px;padding:2px;gap:2px;flex-wrap:wrap}
.seg button{border:0;background:none;padding:5px 11px;border-radius:5px;cursor:pointer;font-size:12.5px;font-weight:500;color:var(--muted)}
.seg button[aria-pressed="true"]{background:var(--surface);color:var(--ink);font-weight:600;box-shadow:var(--shadow)}
.kpi{background:var(--surface);border:1px solid var(--line);border-radius:var(--r);padding:12px 14px;display:flex;flex-direction:column;gap:2px}
.kpi.v{font-family:"Archivo","Public Sans",sans-serif;font-size:24px;font-weight:700;letter-spacing:-.02em;font-variant-numeric:tabular-nums}
.kpi.k{font-size:10.5px;text-transform:uppercase;letter-spacing:.08em;color:var(--muted);font-weight:600}
.kpi.m{font-size:11.5px;color:var(--muted)}
.kpi.crit{border-color:var(--crit)} .kpi.crit.v{color:var(--crit)}
.kpi.warn{border-color:var(--warn)} .kpi.warn.v{color:var(--warn)}
.kpi.ok.v{color:var(--ok)}

/* ----- modal ----- */
.scrim{position:fixed;inset:0;background:rgba(12,18,17,.5);backdrop-filter:blur(2px);z-index:100;display:gri

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    d;grid-template-columns:minmax(0,1fr);place-items:center;padding:16px;overflow-y:auto}
.modal{background:var(--surface);border:1px solid var(--line);border-radius:10px;box-shadow:0 24px 60px rgba
    (0,0,0,.28);width:100%;min-width:0;max-width:min(640px,100%);max-height:92vh;display:flex;flex-direc
    tion:column;margin:auto}
.modal>header{padding:15px 18px;border-bottom:1px solid var(--line);display:flex;align-items:center;gap:10px
    }
.modal>header h3{font-size:15px;font-weight:600;flex:1}
.modal .body{padding:18px;overflow-y:auto;overscroll-behavior:contain}
.modal>footer{padding:12px 18px;border-top:1px solid var(--line);display:flex;gap:8px;justify-content:flex-e
    nd;flex-wrap:wrap;background:var(--surface2);border-radius:0 0 10px 10px}
.modal>footer>*>{min-width:0}
.toast{position:fixed;bottom:18px;left:50%;transform:translateX(-50%);background:var(--ink);color:var(--grou
    nd);padding:10px 16px;border-radius:7px;font-size:13px;font-weight:500;z-index:200;box-shadow:0 8px
    24px rgba(0,0,0,.3);max-width:80vw}

/* ----- availability grid ----- */
.avail{display:grid;grid-template-columns:repeat(7,minmax(0,1fr));gap:6px}
@media (max-width:700px){.avail{grid-template-columns:repeat(4,minmax(0,1fr))}}
.avail .d{border:1px solid var(--line);border-radius:6px;padding:8px 6px;text-align:center;background:var(--
    surface2)}
.avail .d .dn{font-size:10.5px;text-transform:uppercase;letter-spacing:.06em;font-weight:600;color:var(--mut
    ed);margin-bottom:6px}
.avail .d.on{background:var(--accent-soft);border-color:var(--accent)}
.avail .d input[type=time]{font-size:11px;padding:3px 2px;text-align:center;width:100%;min-width:0}
.vmonths{display:grid;grid-template-columns:repeat(auto-fill,minmax(210px,1fr));gap:14px}
.vmon .t{font-weight:700;font-size:12.5px;margin-bottom:5px}
.vgrid{display:grid;grid-template-columns:repeat(7,minmax(0,1fr));gap:2px}
.vgrid span{font-size:9px;color:var(--muted);text-align:center}
.vgrid i{display:block;aspect-ratio:1;border-radius:3px;background:var(--sunken);position:relative;font-styl
    e:normal;font-size:8.5px;color:var(--muted);text-align:center;line-height:2.1}
.vgrid i.h1{background:#e7d9ae}.vgrid i.h2{background:#dfb684}.vgrid i.h3{background:#d99a8f}
.vgrid i.ok{background:var(--accent);color:#fff;font-weight:700}
.vgrid i.hol{outline:2px solid var(--warn);outline-offset:-2px}
@media (prefers-color-scheme: dark){.vgrid i.h1{background:#4d431f}.vgrid i.h2{background:#59391b}.vgrid i.h
    3{background:#5c241c}}
.vleg{display:flex;gap:14px;flex-wrap:wrap;font-size:11.5px;color:var(--muted);margin-top:10px}
.vleg i{width:11px;height:11px;border-radius:3px;display:inline-block;vertical-align:-1px;margin-right:4px}
.avail .d .tog{width:100%;border:0;background:var(--sunken);border-radius:4px;padding:4px;font-size:11px;font
    t-weight:600;cursor:pointer;color:var(--muted);margin-bottom:5px}
.avail .d.on .tog{background:var(--accent);color:var(--on-accent)}

/* ----- week grid ----- */
.week{min-width:900px;border:1px solid var(--line);border-radius:var(--r);overflow:hidden;background:var(--s
    urface)}
.week .hrow,.week .erow{display:grid;grid-template-columns:170px repeat(7,1fr)}
.week .hrow{background:var(--surface2);border-bottom:1px solid var(--line);position:sticky;top:0;z-index:5}
.week .hcell{padding:8px 10px;border-left:1px solid var(--line);font-size:11px}
.week .hcell:first-child{border-left:0}
.week .hcell .dn{font-weight:700;font-family:"Archivo","Public Sans",sans-serif;font-size:12px;text-transfor
    m:uppercase;letter-spacing:.05em}
.week .hcell .dd{color:var(--muted);font-size:11px}
.week .hcell.closed{opacity:.5}

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.week .erow{border-bottom:1px solid var(--line)}
.week .erow:last-child{border-bottom:0}
.week .erow:hover{background:var(--surface2)}
.week .ecell{border-left:1px solid var(--line);padding:5px;min-height:56px;display:flex;flex-direction:column;gap:4px}
.week .name{padding:8px 10px;display:flex;flex-direction:column;gap:2px;justify-content:center;min-width:0}
.week .name .nm{font-weight:600;font-size:13px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
.week .name .meta{font-size:10.5px;color:var(--muted);display:flex;gap:5px;align-items:center;flex-wrap:wrap}
.week .name .bar{height:3px;border-radius:2px;background:var(--sunken);overflow:hidden;margin-top:3px}
.week .name .bar i{display:block;height:100%;background:var(--accent)}
.shiftblk{
  border-radius:5px;padding:4px 6px;font-size:11.5px;line-height:1.3;cursor:pointer;border-left:3px solid var(--accent);
  background:var(--accent-soft);color:var(--accent-ink);position:relative;
}
.shiftblk .lb{font-weight:700;display:block;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
.shiftblk .tm{font-family:"IBM Plex Mono",monospace;font-size:10.5px;opacity:.85;font-variant-numeric:tabular-nums}
.shiftblk.viol{border-left-color:var(--crit);background:var(--crit-soft);color:var(--crit)}
.shiftblk .lk{position:absolute;top:2px;right:4px;font-size:9px;opacity:.7}
.shiftblk.chg::before{content:"";position:absolute;top:3px;left:3px;width:6px;height:6px;border-radius:50%;background:var(--warn)}
.card.folded>*.not(header){display:none}
.fixopt.bestfix{border-color:var(--ok);background:color-mix(in srgb,var(--ok) 5%,var(--surface))}
.hnotice{display:flex;align-items:center;gap:10px;justify-content:space-between;
  border-left:3px solid var(--accent);background:var(--surface);border-radius:0 7px 7px 0;
  padding:9px 12px;font-size:13px;color:var(--ink2)}
.mgrid{display:grid;grid-template-columns:repeat(7,minmax(0,1fr));gap:4px}
.mgrid.mhead{margin-bottom:4px;font-size:10.5px;color:var(--muted);text-transform:uppercase;letter-spacing:.05em}
.mgrid.mhead div{text-align:center}
.mcell{position:relative;border:1px solid var(--line);border-radius:7px;background:var(--surface);min-height:58px;
  display:flex;flex-direction:column;align-items:flex-start;gap:2px;padding:6px 7px;font:inherit;color:inherit;cursor:pointer;text-align:left}
.mcell:hover{border-color:var(--line-strong)}
.mcell.off{border:0;background:transparent;cursor:default}
.mcell.closed{background:var(--sunken);color:var(--muted)}
.mcell.posted{border-color:var(--ok);background:var(--ok-soft)}
.mcell.draft{border-color:var(--accent);background:var(--accent-soft)}
.mcell.today .md{background:var(--accent);color:var(--on-accent);border-radius:99px;padding:0 6px;margin-left:-4px}
.mcell .md{font-weight:700;font-size:12.5px}
.mcell .mn{font-size:10.5px;color:var(--muted)}
.mcell .mn.faint{color:var(--faint)}
.mcell .mh{position:absolute;top:5px;right:7px;font-size:10px;color:var(--warn)}
@media (max-width:820px){ .mcell{min-height:48px;padding:4px 5px} }
.shiftblk.cout,.drow.cout{border-style:dashed;opacity:.75}
.shiftblk.cout .lb::after,.drow.cout .lb::after{content:" · out";font-weight:600;color:var(--crit)}
.drow{position:relative}
.drow.chg::before{content:"";position:absolute;top:6px;left:-2px;width:6px;height:6px;border-radius:50%;back

```

```

        ground:var(--warn))}
.addbtn{border:1px dashed var(--line2);background:none;border-radius:5px;color:var(--faint);cursor:pointer;font-size:15px;line-height:1;padding:3px 0;margin-top:auto}
.addbtn: hover{border-color:var(--accent);color:var(--accent);background:var(--accent-soft)}
.week .ecell.off{background:repeating-linear-gradient(45deg,transparent,transparent 5px,var(--sunken) 5px,var(--sunken) 6px)}

/* ----- coverage bar ----- */
.stblk{display:flex;gap:6px;align-items:center;margin-bottom:3px;font-size:12px;white-space:nowrap}
.stblk .mono{color:var(--muted);font-size:11px;min-width:88px}
.covbar{display:flex;height:26px;border-radius:5px;overflow:hidden;border:1px solid var(--line);background:var(--sunken)}
.covbar i{flex:1;border-right:1px solid var(--surface);position:relative}
.covbar i:last-child{border-right:0}
.covbar i.g{background:var(--ok-soft)} .covbar i.y{background:var(--warn-soft)} .covbar i.r{background:var(--crit-soft)} .covbar i.n{background:transparent}
.covbar i.shut{background:repeating-linear-gradient(45deg,var(--sunken),var(--sunken) 3px,var(--line) 3px,var(--line) 4px)}
.covscale{display:flex;font-size:9.5px;color:var(--faint);font-family:"IBM Plex Mono",monospace;margin-top:3px}
.covscale span{flex:1;text-align:left}

/* ----- day timeline ----- */
.tl{min-width:820px}
.tlrow{display:grid;grid-template-columns:160px 1fr;gap:10px;align-items:center;padding:6px 0;border-bottom:1px solid var(--line)}
.tlrow:last-child{border-bottom:0}
.tlname{font-size:13px;font-weight:600;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
.tlname small{display:block;font-weight:400;color:var(--muted);font-size:11px}
.tltrack{position:relative;height:34px;background:var(--sunken);border-radius:5px}
.tlshut{position:absolute;top:0;bottom:0;background:repeating-linear-gradient(45deg,transparent,transparent 4px,var(--line2) 4px,var(--line2) 5px);opacity:.55;pointer-events:auto}
.tlbar{position:absolute;top:0;bottom:0;background:var(--accent-soft);border:1px solid var(--accent);border-radius:5px;display:flex;align-items:center;padding:0 6px;font-size:11px;color:var(--accent-ink);font-weight:600;overflow:hidden;white-space:nowrap}
.tlmeal{position:absolute;top:2px;bottom:2px;background:var(--warn);border-radius:3px;display:grid;place-items:center;color:var(--on-warn);font-size:9px;font-weight:700;overflow:hidden;z-index:2}
.tlmeal.apart{outline:2px dashed var(--ink2);outline-offset:1px}
.tlrest{position:absolute;top:2px;bottom:2px;background:var(--ok);border-radius:2px;z-index:2;min-width:4px}
.tlhead{display:grid;grid-template-columns:160px 1fr;gap:10px;padding-bottom:4px}
.tlticks{position:relative;height:16px;font-family:"IBM Plex Mono",monospace;font-size:9.5px;color:var(--faint)}
.tlticks span{position:absolute;transform:translateX(-50%);white-space:nowrap}
.tlgrid{position:absolute;inset:0;pointer-events:none}
.tlgrid i{position:absolute;top:0;bottom:0;width:1px;background:var(--line)}

/* ----- misc ----- */
.viollist{display:flex;flex-direction:column;gap:0}
.viol-i{display:flex;gap:10px;padding:9px 14px;border-bottom:1px solid var(--line);align-items:flex-start;font-size:13px}
.viol-i:last-child{border-bottom:0}
.viol-i .sv{width:3px;align-self:stretch;border-radius:2px;flex:none}

```

```

.viol-i .sv.crit{background:var(--crit)} .viol-i .sv.warn{background:var(--warn)} .viol-i .sv.info{background:var(--muted)}
.viol-i .tx strong{display:block;font-weight:600;font-size:13px}
.viol-i .tx span{color:var(--muted);font-size:12px}
.rank{display:flex;flex-direction:column;gap:6px}
.rankitem{display:flex;gap:10px;align-items:center;padding:8px 10px;border:1px solid var(--line);border-radius:6px;background:var(--surface2)}
.rankitem.no{width:22px;height:22px;border-radius:5px;background:var(--accent);color:var(--on-accent);display:grid;place-items:center;font-size:11px;font-weight:700;flex:none}
.codebox{font-family:"IBM Plex Mono",monospace;font-size:11px;background:var(--sunken);border:1px solid var(--line);border-radius:6px;padding:10px;word-break:break-all;max-height:130px;overflow:auto;line-height:1.45}
.banner{display:flex;flex-wrap:wrap;gap:10px;padding:11px 14px;border-radius:var(--r);font-size:13px;align-items:flex-start;border:1px solid}
.banner>*[min-width:0;max-width:100%].banner>.btn{white-space:normal;text-align:left}
/* on a narrow phone a banner's action goes full width rather than pushing the page wide */
@media (max-width:430px){ .banner>.btn{width:100%;margin-left:0!important;justify-content:center} }
.banner.info{background:var(--accent-soft);border-color:transparent;color:var(--accent-ink)}
.banner.warn{background:var(--warn-soft);border-color:transparent;color:var(--warn)}
.banner.crit{background:var(--crit-soft);border-color:transparent;color:var(--crit)}
.banner b{font-weight:700}
.legend{display:flex;gap:14px;flex-wrap:wrap;font-size:11.5px;color:var(--muted);align-items:center}
.legend i{display:inline-block;width:10px;height:10px;border-radius:3px;margin-right:5px;vertical-align:-1px}
.mini{font-size:11px;color:var(--muted)}
.ovf{max-height:280px;overflow-y:auto}

/* shift + role colours that survive both themes */
.sc{color:var(--sc)}
.shiftblk[style*="--sc"],.tlbar[style*="--sc"]{background:color-mix(in srgb,var(--sc) 12%,transparent);border-color:var(--sc);color:color-mix(in srgb,var(--sc) 88%,var(--ink))}
.tag[style*="--sc"]{background:color-mix(in srgb,var(--sc) 15%,transparent);color:color-mix(in srgb,var(--sc) 86%,var(--ink))}
.dotc{display:inline-block;width:8px;height:8px;border-radius:2px;background:var(--sc);margin-right:5px;vertical-align:-1px}
.shiftblk.viol[style*="--sc"]{background:var(--crit-soft);color:var(--crit);border-color:var(--crit);border-left-color:var(--crit)}
@media (prefers-color-scheme:dark){
  :root:not([data-theme="light"]) .shiftblk[style*="--sc"],:root:not([data-theme="light"]) .tlbar[style*="--sc"]{background:color-mix(in srgb,var(--sc) 24%,transparent);color:color-mix(in srgb,var(--sc) 40%,#ffffff)}
  :root:not([data-theme="light"]) .tag[style*="--sc"]{background:color-mix(in srgb,var(--sc) 26%,transparent);color:color-mix(in srgb,var(--sc) 38%,#ffffff)}
}
:root[data-theme="dark"] .shiftblk[style*="--sc"],:root[data-theme="dark"] .tlbar[style*="--sc"]{background:color-mix(in srgb,var(--sc) 24%,transparent);color:color-mix(in srgb,var(--sc) 40%,#ffffff)}
:root[data-theme="dark"] .tag[style*="--sc"]{background:color-mix(in srgb,var(--sc) 26%,transparent);color:color-mix(in srgb,var(--sc) 38%,#ffffff)}
.week .ecell .addbtn{opacity:0}
.week .ecell:hover .addbtn,.week .ecell .addbtn:focus-visible{opacity:1}
@media (hover:none){.week .ecell .addbtn{opacity:.6}}

```

```

.setbody{padding:4px 0}
.setrow{display:flex;gap:16px;align-items:center;padding:11px 16px;border-bottom:1px solid var(--line);flex-wrap:wrap}
.snote{border-bottom:1px dashed var(--line2);background:transparent;font-size:11.5px;color:var(--muted);width:100%;max-width:210px;padding:2px 0;margin-top:3px}
.snote:focus{outline:0;border-bottom-color:var(--acc);color:var(--fg)}
.devrow{display:flex;gap:10px;align-items:center;padding:9px 0;border-bottom:1px solid var(--line);flex-wrap:wrap}
.devrow:last-child{border-bottom:0}
.setrow:last-child{border-bottom:0}
.setrow .sq{flex:1;min-width:220px}
.setrow .sq strong{font-size:13.5px;font-weight:600;display:block}
.setrow .sq .mini{max-width:64ch;margin-top:2px}
.setrow .sc{display:flex;gap:8px;align-items:center;flex:none}
.setrow .sc select,.setrow .sc input{min-width:150px}
@media (max-width:640px){.setrow .sc{width:100%}.setrow .sc select,.setrow .sc input{flex:1;min-width:0}}

#collabbar{position:fixed;bottom:0;left:0;right:0;z-index:150;background:var(--ink);color:var(--ground);padding:10px 16px;display:flex;gap:12px;align-items:center;justify-content:center;flex-wrap:wrap;font-size:13px;box-shadow:0 -4px 20px rgba(0,0,0,.25)}
#collabbar .btn{background:var(--surface);color:var(--ink);border-color:var(--line2)}

.busy,.quiet{font-size:9.5px;font-weight:700;text-transform:uppercase;letter-spacing:.05em;padding:1px 5px;border-radius:99px;margin-left:4px;vertical-align:1px}
.busy{background:var(--warn-soft);color:var(--warn)}
.quiet{background:var(--sunken);color:var(--muted)}

.tup{display:inline-block;background:var(--accent);color:var(--on-accent);border-radius:4px;font-size:9px;font-weight:800;line-height:1;padding:2px 4px;vertical-align:1px;margin-left:3px}
.legend .tup{margin:0 4px 0 0}

.wklist{display:flex;flex-direction:column}
.wkrow{display:flex;gap:16px;align-items:center;padding:13px 16px;border-bottom:1px solid var(--line);cursor:pointer;transition:background .12s;flex-wrap:wrap;text-align:left}
.wkrow *{text-align:left}
.wkrow:last-child{border-bottom:0}
.wkrow:hover{background:var(--surface2)}
.wkrow.here{box-shadow:inset 3px 0 0 var(--accent)}
.wkrow.posted{opacity:.86}
.wkwhen{min-width:190px;flex:1}
.wkwhen strong{font-family:"Archivo","Public Sans",sans-serif;font-size:14px;letter-spacing:-.01em}
.wknote{font-size:12px;color:var(--accent-ink);background:var(--accent-soft);border-radius:5px;padding:3px 7px;margin-top:5px;display:inline-block}
.wkstate{width:110px;flex:none}
.wkfacts{display:flex;gap:12px;align-items:center;flex-wrap:wrap;flex:1.4;min-width:180px;font-size:12.5px}
.wkacts{display:flex;gap:6px;flex:none}
@media (max-width:820px){.wkstate,.wkfacts{width:auto;flex:1 1 100%}.wkacts{flex:1 1 100%}}

.abar{flex:1;height:7px;border-radius:4px;background:var(--sunken);overflow:hidden;min-width:60px}

```

- Intervening pages omitted -

Source pages 11 through 268 are not included in this deposit.

```

/* =====
Labor cost
===== */
/* Holidays this week, and who gets paid for them whether they work or not. */
function holidaysIn(ws){
  const dates=weekDates(ws).map(d=>d.date);
  return (S.holidays.list||[]).filter(h=>dates.includes(h.date));
}
function holidayPayFor(e, ws){
  if(!e.holidayPay) return {hours:0, days:0};
  const n=holidaysIn(ws).length;
  return {hours: n*(e.holidayPayH||8), days:n};
}
function holidayPayCost(ws){
  if(!costOn()) return 0;
  return S.employees.filter(x=>x.active).reduce((t,e)=>{
    const hp=holidayPayFor(e,ws);
    return t + hp.hours*(e.rate||0);
  },0);
}
function costOf(a){ const e=empById(a.empId); if(!e||!e.rate) return 0;
  const paid=paidMinutes(a)/60;
  const ot=Math.max(0,paid-(S.org.otDailyH||8)), reg=paid-ot;
  return reg*e.rate + ot*e.rate*1.5;
}
function weekCost(ws){ return peekWeek(ws).assignments.reduce((t,a)=>t+costOf(a),0) + holidayPayCost(ws); }
function money(n){ return "$"+Math.round(n).toLocaleString(); }
function haveRates(){ return !!S.org.trackCost && S.employees.some(e=>e.rate>0); }
function costOn(){ return !!S.org.trackCost; }

/* =====
Week templates
===== */
function saveTemplate(){
  const wk=getWeek();
  if(!wk.assignments.length){ toast("Nothing to save – build the week first"); return; }
  modal("Save this week as a template",
    `

{
      S.templates = S.templates||[];
      S.templates.push({id:uid("tpl"), name:r.querySelector("#tn").value.trim()||"Template",
        rows: wk.assignments.map(a=>({empId:a.empId,dow:a.dow,shiftId:a.shiftId,roleId:a.roleId})), at:Date.
        now()});
      save(); closeModal(); render(); toast("Template saved");
    }; });
}
function applyTemplate(id){


```

```

const t=(S.templates||[]).find(x=>x.id===id); if(!t) return;
const wk=getWeek(); const dates=weekDates();
wk.assignments = wk.assignments.filter(a=>a.locked);
t.rows.forEach(r=>{
  const d = dates.find(x=>x.dow===r.dow); if(!d) return;
  const sh = shiftById(r.shiftId); if(!sh) return;
  if(!empById(r.empId)) return;
  if(wk.assignments.some(a=>a.empId===r.empId && a.date===d.date)) return;
  const [aM,bM]=shiftMinutes(sh);
  const rec={id:uid("a"),empId:r.empId,date:d.date,dow:r.dow,shiftId:r.shiftId,startMin:aM,endMin:bM,roleId:r.roleId,locked:false};
  applyBreaks(rec); wk.assignments.push(rec);
});
dates.forEach(({date})=>staggerDay(wk.assignments.filter(a=>a.date===date && !a.locked), wk.assignments.filter(a=>a.date===date)));
wk.assignments.sort((x,y)>=>x.date.localeCompare(y.date)||x.startMin-y.startMin);
wk.generatedAt=Date.now(); save(); render(); toast("Template applied");
}

/* =====
Command palette + shortcuts
===== */
function palette(){
  if($("#palroot")) return;
  const items=[];
  NAV.forEach(n=>{ if(n.id) items.push({t:"Go to "+n.label, kind:"view", run:()=>go(n.id)}); });
  items.push({t:"Build this week's schedule", kind:"action", run:()=>{ const r=generateSchedule(); save(); go("week"); toast(r.filled+" shifts placed"); }});
  items.push({t:"Add a person", kind:"action", run:()=>{ go("team"); setTimeout(()=>empModal(null),60); }});
  items.push({t:"Add a shift", kind:"action", run:()=>{ go("shifts"); setTimeout(()=>shiftModal(null),60); }});
  items.push({t:"Log time off", kind:"action", run:()=>{ go("team"); setTimeout(()=>toffModal(),60); }});
  items.push({t:"Change who gets first pick of shifts", kind:"view", run:()=>go("priority")});
  items.push({t:"Redo this week – I don't like this draft", kind:"action", run:()=>redoModal(S.currentWeek)});
  items.push({t:"Report a problem", kind:"action", run:()=>bugModal()});
  items.push({t:(advanced()?"Switch to the simple view":"Show the advanced setup pages"), kind:"action", run:()=>{ S.ui=S.ui||{}; S.ui.advanced=!advanced(); save(); render(); }});
  items.push({t:"Settings – every setup answer", kind:"view", run:()=>go("settings")});
  items.push({t:"Analytics – hours, trends, denied requests", kind:"view", run:()=>go("analytics")});
  items.push({t:"Run setup again", kind:"action", run:()=>wizard()});
  items.push({t:"Add a scheduling request", kind:"view", run:()=>go("requests")});
  items.push({t:"All schedules – drafts and posted weeks", kind:"view", run:()=>go("weeks")});
  items.push({t:(S.rules.fillMode==="full"? "Schedule only the cover I asked for": "Fully schedule everyone's contracted hours"), kind:"action", run:()=>{ S.rules.fillMode = S.rules.fillMode==="full"? "demand":"full"; save(); const r=generateSchedule(); render(); toast(r.filled+" shifts drafted"); }});
  items.push({t:"Cover the open shifts", kind:"action", run:()=>fixModal(S.currentWeek)});
  items.push({t:"Rotate this week's stations", kind:"action", run:()=>{ assignStationsWeek(S.currentWeek); render(); toast("Stations handed out again"); }});
  items.push({t:"Decide which request to decline", kind:"action", run:()=>requestConflictModal(S.currentWeek)});
}

```

```

items.push({t:"Change what shape this week runs", kind:"action", run:()=>weekPatternModal(S.currentWeek)})
;
items.push({t:"Build a week pattern", kind:"action", run:()=>{ go("demand"); setTimeout(patternModal,80);
}});
items.push({t:"Draft the schedule", kind:"action", run:()=>{ go("week"); setTimeout(draftModal,80); }});
items.push({t:(S.org.trackCost?"Turn off":"Turn on")+" labor cost tracking", kind:"action",
run:()=>{ S.org.trackCost=!S.org.trackCost; save(); render(); toast("Labor cost "+(S.org.trackCost?"on":
"off")); }});
PRI_MODES.forEach(m=>items.push({t:"Set shift priority to: "+m[1], kind:"action",
run:()=>{ S.priority.mode=m[0]; if(m[0]==="custom"&&!(S.priority.order||[]).length) S.priority.order=pri
orityOrder();
priorityRefresh(); save(); go("priority"); toast("Priority set to "+m[1].toLowerCase()); }));
items.push({t:"Weekly schedule PDF", kind:"export", run:()=>exportWeekPDF()});
items.push({t:"Week grid for Excel", kind:"export", run:()=>saveFile("schedule-grid-"+S.currentWeek+".csv"
, weekGridCSV())});
items.push({t:"Next week", kind:"action", run:()=>{ S.currentWeek=addDays(S.currentWeek,7); save(); render
(); }});
items.push({t:"Previous week", kind:"action", run:()=>{ S.currentWeek=addDays(S.currentWeek,-7); save(); r
ender(); }});
items.push({t:"Undo the last change", kind:"action", run:()=>undo()});
S.employees.forEach(e=>items.push({t:e.name, kind:"person", run:()=>{ go("team"); setTimeout(()=>empModal(
e.id),60); }}));
DAYS.forEach((d,i)>items.push({t:d+" day sheet", kind:"view", run:()=>{ DAYVIEW=i; go("day"); }}));

const root=document.createElement("div"); root.id="palroot";
root.innerHTML=`<div class="pal"><div class="palbox">
  <input type="text" id="palq" placeholder="Jump to a view, a person, or an action..." autocomplete="off">
  <div class="palres" id="palr"></div>
  <div class="palfoot"><span><kbd>↑</kbd><kbd>↓</kbd> move</span><span><kbd>⌵</kbd> run</span><span><kbd>e
sc</kbd> close</span></div>
</div></div>`;
document.body.appendChild(root);
const q=root.querySelector("#palq"), res=root.querySelector("#palr");
let sel=0, shown=items;
const draw=()=>{
  const v=q.value.toLowerCase().trim();
  shown = v? items.filter(i=>i.t.toLowerCase().includes(v)) : items;
  sel=clamp(sel,0,Math.max(0,shown.length-1));
  res.innerHTML = shown.length? shown.slice(0,12).map((i,n)>
    `<button data-n="${n}" data-sel="${n===sel?1:0}">${esc(i.t)}<span class="kind">${i.kind}</span></button>
    `).join("")
    : `<div class="empty" style="padding:20px">Nothing matches "${esc(q.value)}".</div>`;
  res.querySelectorAll("button").forEach(b=>b.onclick=()=>{ close(); shown[+b.dataset.n].run(); });
};
const close=()=>{ root.remove(); document.removeEventListener("keydown",key,true); };
const key=ev=>{
  if(ev.key==="Escape"){ ev.stopPropagation(); close(); }
  else if(ev.key==="ArrowDown"){ ev.preventDefault(); sel=Math.min(sel+1,Math.min(11,shown.length-1)); dra
w(); }
  else if(ev.key==="ArrowUp"){ ev.preventDefault(); sel=Math.max(0,sel-1); draw(); }
  else if(ev.key==="Enter"){ ev.preventDefault(); if(shown[sel]){ const r=shown[sel].run; close(); r(); }
  }
}

```

```

};
document.addEventListener("keydown",key,true);
root.querySelector(".pal").addEventListener("mousedown",e=>{ if(e.target.classList.contains("pal")) close(
); });
q.oninput={()=>{ sel=0; draw(); }; draw(); q.focus();
}
document.addEventListener("keydown",e=>{
const typing = /^(INPUT|TEXTAREA|SELECT)$/.test((e.target.tagName||""));
if((e.metaKey||e.ctrlKey) && e.key.toLowerCase()=== "k"){ e.preventDefault(); palette(); return; }
if((e.metaKey||e.ctrlKey) && e.key.toLowerCase()=== "z" && !typing){ e.preventDefault(); undo(); return; }
if(typing || e.metaKey || e.ctrlKey || e.altKey) return;
if($("#modalroot").innerHTML || $("#palroot")) return;
if(e.key=== "/"){ e.preventDefault(); palette(); }
else if(e.key=== "w"){ go("week"); } else if(e.key=== "d"){ go("day"); }
else if(e.key=== "t"){ go("team"); } else if(e.key=== "o"){ go("home"); }
else if(e.key=== "h"){ go("holidays"); } else if(e.key=== "e"){ go("export"); }
else if(e.key=== "r"){ go("requests"); } else if(e.key=== ","){ go("settings"); }
else if(e.key=== "ArrowLeft" && (VIEW=== "week" || VIEW=== "overview")){ S.currentWeek=addDays(S.currentWeek, -7
); save(); render(); }
else if(e.key=== "ArrowRight" && (VIEW=== "week" || VIEW=== "overview")){ S.currentWeek=addDays(S.currentWeek, 7
); save(); render(); }
});

/* =====
Per-person schedule cards
===== */
async function exportPersonCards(){
if(!jsPDFReady()){ pdfUnavailable(); return; }
const ws=S.currentWeek, wk=getWeek(ws), dates=weekDates(ws), L=law();
const doc=newDoc("portrait");
const W=doc.internal.pageSize.getWidth();
const people=S.employees.filter(e=>e.active && wk.assignments.some(a=>a.empId===e.id));
if(!people.length){ toast("Nothing scheduled to hand out yet"); return; }
let y=52, first=true;
people.forEach((e,idx)=>{
if(!first && y>620){ doc.addPage(); y=52; }
if(first){ pdfHeader(doc,(S.org.name||"Schedule"),"Individual schedules · week of "+prettyFull(ws)); y=8
8; first=false; }
doc.setFont("helvetica","bold"); doc.setFontSize(11); doc.setTextColor(20,32,30);
doc.text(e.name, 40, y);
doc.setFont("helvetica","normal"); doc.setFontSize(8); doc.setTextColor(105,119,115);
doc.text(fmtHrs(empWeekMinutes(e.id,ws))+" · "+e.roleIds.map(roleAbbr).join("/"), W-40, y, {align:"right
"});
doc.autoTable({
body: dates.map(d=>{
const mine=wk.assignments.filter(a=>a.empId===e.id && a.date===d.date);
if(!mine.length) return [DSHORT[d.dow]+" "+prettyDate(d.date), onTimeOff(e.id,d.date)? ((timeOffFor(
e.id,d.date)||{}).type||"Off").toUpperCase() : "Off", "", ""];
const a=mine[0], b=a.breaks||{meals:[],rests:[]};
return [DSHORT[d.dow]+" "+prettyDate(d.date), (shiftById(a.shiftId)||{label:""}).label,
m2disp(a.startMin)+" - "+m2disp(a.endMin),
(S.rules.breakMode=== "assigned" && b.meals.length? "meal "+b.meals.filter(m=>!m.waived).map(m=>m2dis

```

```

    p(m.start)).join("/") : b.owed.meals? b.owed.meals+" meal owed":""))+
    (S.rules.restMode=="assigned"&&b.rests.length? "  · rests "+b.rests.map(r=>m2disp(r.start)).join
    ("/") : b.owed.rests? "  · "+b.owed.rests+" rest owed":""));
  }},
  startY:y+7, theme:"grid",
  styles:{fontSize:7.6,cellPadding:4,lineColor:[222,228,226],lineWidth:.5,textColor:[45,58,55]},
  columnStyles:{0:{cellWidth:76,fontStyle:"bold"},1:{cellWidth:74},2:{cellWidth:88}},
  margin:{left:40,right:40}
});
y = doc.lastAutoTable.finalY + 26;
if(y>690 && idx<people.length-1){ doc.addPage(); y=52; }
});
pdfFooter(doc);
await saveFile("individual-schedules-"+ws+".pdf", doc.output("arraybuffer"));
}

```

```

/* =====
Week: more menu, open-shift filler, drag and drop
===== */
function swapModal(ws){
  ws=ws||S.currentWeek;
  const wk=getWeek(ws);
  if(wk.assignments.length<2){ toast("Two shifts on the board are needed for a trade"); return; }
  const opt=a=>{ const e=empById(a.empId)||{name:"-"}, sh=aShift(a);
    return `<option value="${a.id}">${DSHORT[a.dow]} · ${m2disp(a.startMin)}-${m2disp(a.endMin)} ${esc(sh.la
    bel)} · ${esc(e.name)}</option>`; };
  const sorted=wk.assignments.slice().sort((a,b)=>a.date.localeCompare(b.date)||a.startMin-b.startMin);
  modal("Check a trade", `
    <div class="stack">
      <div class="help">Two people want to swap. Pick both shifts and the trade is checked against every rul
      e – hours, turnaround, availability, coverage, licenses – before anything moves. Nothing changes unt
      il you say so.</div>
      <div class="field"><label>First shift</label><select id="sw1">${sorted.map(opt).join("")}</select></di
      v>
      <div class="field"><label>Trades with</label><select id="sw2">${sorted.map(opt).join("")}</select></di
      v>
      <div id="swout" class="stack"></div>
    </div>`,
    `<button class="btn" data-close>Cancel</button><button class="btn" id="swchk">Check it</button><button c
    lass="btn primary" id="swgo" disabled>Make the trade</button>`,
    r=>{
      let checked=null;
      const box=r.querySelector("#swout"), go=r.querySelector("#swgo");
      const reset=()=>{ checked=null; go.disabled=true; box.innerHTML=""; };
      r.querySelector("#sw1").onchange=reset; r.querySelector("#sw2").onchange=reset;
      const doSwap=(x,y)=>{ const t=x.empId; x.empId=y.empId; y.empId=t;
        const ex=empById(x.empId), ey=empById(y.empId);
        if(x.roleId && ex && !ex.roleIds.includes(x.roleId)) x.roleId=ex.roleIds[0]||null;
        if(y.roleId && ey && !ey.roleIds.includes(y.roleId)) y.roleId=ey.roleIds[0]||null;
        applyBreaks(x); applyBreaks(y); };
      r.querySelector("#swchk").onclick=()=>{

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const a1=wk.assignments.find(a=>a.id===r.querySelector("#sw1").value);
const a2=wk.assignments.find(a=>a.id===r.querySelector("#sw2").value);
if(!a1||!a2||a1===a2){ box.innerHTML='<div class="banner warn"><div>Pick two different shifts.</div>
</div>'; return; }
if(a1.empId===a2.empId){ box.innerHTML='<div class="banner warn"><div>Those are both '+esc((empById(
a1.empId)|{name:"the same person"}).name)+"'s shifts – nothing to trade.</div></div>"; return; }
const e1=empById(a1.empId), e2=empById(a2.empId);
if(a1.roleId && e2 && !e2.roleIds.includes(a1.roleId)){
  box.innerHTML='<div class="banner warn"><div><b>'+esc(e2.name)+"</b> isn't a "+esc(roleName(a1.roleId).toLowerCase()+"", which that shift requires. The trade can't work as-is.</div></div>"; return;
}
if(a2.roleId && e1 && !e1.roleIds.includes(a2.roleId)){
  box.innerHTML='<div class="banner warn"><div><b>'+esc(e1.name)+"</b> isn't a "+esc(roleName(a2.roleId).toLowerCase()+"", which that shift requires. The trade can't work as-is.</div></div>"; return;
}
const before=new Set(validate(ws).map(vKey));
const keep1={empId:a1.empId,roleId:a1.roleId}, keep2={empId:a2.empId,roleId:a2.roleId};
doSwap(a1,a2);
const fresh=validate(ws).filter(v=>!before.has(vKey(v)));
/* put it back exactly */
a1.empId=keep1.empId; a1.roleId=keep1.roleId; a2.empId=keep2.empId; a2.roleId=keep2.roleId;
applyBreaks(a1); applyBreaks(a2);
if(!fresh.length){
  box.innerHTML='<div class="banner info"><div><b>Clean trade.</b> Every rule still holds with '+esc(
(e1.name.split(" ")[0])+" and "+esc(e2.name.split(" ")[0])+" swapped.</div></div>";
  checked={a1,a2}; go.disabled=false;
} else {
  box.innerHTML='<div class="banner warn"><div><b>This trade breaks '+fresh.length+' thing'+(fresh.length===1?"":"s")+':</b><br>'+
    fresh.slice(0,4).map(v=>esc(v.title)+" – "+esc(v.detail)).join("<br>")+
    (fresh.length>4? "<br>...and "+(fresh.length-4)+" more":"")+
    "</div></div>"+'<div class="mini">You can still make it if the trade matters more – it stays flagged in the conflicts list.</div>';
  checked={a1,a2}; go.disabled=false;
}
};
go.onclick=()=>{
  if(!checked) return;
  doSwap(checked.a1, checked.a2);
  checked.a1.locked=true; checked.a2.locked=true;
  save(); closeModal(); render(); toast("Traded and locked");
};
});
}
function weekMoreModal(){
  const tpls = S.templates|[];
  modal("Week tools", `
    <div class="stack">
      <div class="stack" style="gap:8px">
        <div class="lab">This week</div>
        ${sendOn()? `<button class="btn primary" id="msend">Send it to the team</button>`: ""}
        <button class="btn" id="mclear">Clear everything that isn't locked</button>
      </div>
    </div>
  `);
}

```

```

    <button class="btn" id="mcopy">Copy last week forward</button>
    <button class="btn" id="mswap">Check a trade – two people swapping shifts</button>
    <button class="btn" id="munlock">Unlock every shift</button>
  </div>
  <div class="sep"></div>
  <div class="stack" style="gap:8px">
    <div class="lab">Templates</div>
    <button class="btn" id="msave">Save this week as a template</button>
    ${tpls.length? tpls.map(t=>`<div class="row" style="justify-content:space-between;border:1px solid v
ar(--line);border-radius:6px;padding:7px 10px">
      <div><strong>${esc(t.name)}</strong><div class="mini">${t.rows.length} shifts · saved ${new Date(t
.at).toLocaleDateString()}</div></div>`)}
    <div class="row" style="gap:6px"><button class="btn sm primary" data-apptpl="${t.id}">Apply</butto
n>
      <button class="btn sm danger" data-deltpl="${t.id}">Delete</button></div></div>`).join("")
    : `<div class="mini">No templates yet. Build a week you like, then save it – applying it later tak
es one click.</div>`}
  </div>
  <div class="sep"></div>
  <div class="stack" style="gap:8px">
    <div class="lab">Hand out</div>
    <button class="btn" id="mcards">Individual schedules, one card each (PDF)</button>
  </div>
  <div class="sep"></div>
  <div class="stack" style="gap:8px">
    <div class="lab">Gaps</div>
    <button class="btn" id="mfix">Cover the open shifts</button>
    <button class="btn" id="mdeny">Decide which request to decline</button>
  </div>
</div>`, `<button class="btn" data-close>Close</button>`,
r=>{
  const sb=r.querySelector("#msend"); if(sb) sb.onclick=()=>{ closeModal(); sendModal(S.currentWeek); };
  const swb=r.querySelector("#mswap"); if(swb) swb.onclick=()=>{ closeModal(); swapModal(S.currentWeek);
  };
  r.querySelector("#mclear").onclick=()=>{ const wk=getWeek(); wk.assignments=wk.assignments.filter(a=>a
.locked); wk.unfilled=[]; wk.touched=true; wk.suggested=false; wk.auto=false; save(); closeModal();
  render(); };
  r.querySelector("#munlock").onclick=()=>{ getWeek().assignments.forEach(a=>a.locked=false); save(); cl
oseModal(); render(); toast("All shifts unlocked"); };
  r.querySelector("#mcopy").onclick=()=>{
    const prev=S.weeks[addDays(S.currentWeek,-7)];
    if(!prev||!prev.assignments.length){ toast("Nothing to copy from last week"); return; }
    const wk=getWeek();
    wk.assignments = prev.assignments.map(a=>{
      const c=Object.assign({},a,{id:uid("a"),date:addDays(a.date,7),locked:false});
      applyBreaks(c); return c;
    });
    wk.generatedAt=Date.now(); wk.published=false; save(); closeModal(); render(); toast("Copied last we
ek forward");
  };
  r.querySelector("#msave").onclick=()=>{ closeModal(); saveTemplate(); };
  r.querySelector("#mcards").onclick=()=>{ closeModal(); exportPersonCards(); };

```

```

    r.querySelector("#mfix").onclick={()=>{ closeModal(); fixModal(S.currentWeek); }};
    r.querySelector("#mdeny").onclick={()=>{ closeModal(); requestConflictModal(S.currentWeek); }};
    r.querySelectorAll("[data-apptpl]").forEach(b=>b.onclick={()=>{ closeModal(); applyTemplate(b.dataset.a
      pptpl); }});
    r.querySelectorAll("[data-deltpl]").forEach(b=>b.onclick={()=>{ S.templates=(S.templates||[]).filter(t=
      >t.id!=b.dataset.deltpl); save(); closeModal(); render(); }});
  });
}

function fillOpenModal(date, shiftId, roleId){
  const dow=parseYmd(date).getDay();
  const sh=shiftById(shiftId)||{label:"Shift"};
  const [aMin,bMin]=shiftMinutes(sh);
  const cand=candidatesFor(shiftId,date,dow,roleId,null);
  const free=cands.filter(c=>!c.blocks.length), blocked=cands.filter(c=>c.blocks.length);
  modal("Who could work "+sh.label+"?", `
    <div class="stack">
      <div class="mini">${DAYS[dow]} ${prettyFull(date)} · ${m2disp(aMin)}-${m2disp(bMin)} · ${esc(roleName(
        roleId))}</div>
      ${free.length? `<div><div class="lab" style="margin-bottom:6px">Clear to take it</div>${candidateHTML(
        free,"data-pick")}</div>`
        : `<div class="banner warn"><div><b>Nobody clears every limit.</b> The list below shows exactly what
          is in the way for each person – sometimes the answer is to move a different shift first, or to acce
          pt one overtime hour.</div></div>`}
      ${blocked.length? `<div><div class="lab" style="margin:12px 0 6px">Blocked, and by what</div>${candida
        teHTML(blocked,"data-pick")}</div>`.":""}
    </div>`, `<button class="btn" data-close>Close</button>`,
    r=>{
      r.querySelectorAll("[data-pick]").forEach(b=>b.onclick={()=>{
        const empId=b.dataset.pick, e=empById(empId);
        const rec={id:uid("a"),empId,date,dow,shiftId,startMin:aMin,endMin:bMin,
          roleId:roleId||e.roleIds[0]||null,locked:true};
        applyBreaks(rec);
        const wk=getWeek(weekStartOf(date,S.org.weekStart));
        wk.assignments.push(rec);
        wk.assignments.sort((x,y)=>x.date.localeCompare(y.date)||x.startMin-y.startMin);
        staggerDay(wk.assignments.filter(a=>a.date===date && !a.locked), wk.assignments.filter(a=>a.date===d
          ate));
        save(); closeModal(); render(); toast(e.name+" added and locked");
      }});
    });
}

let DRAG=null;
function wireDragDrop(root){
  root.querySelectorAll(".shiftblk[data-edita]").forEach(el=>{
    el.setAttribute("draggable","true");
    el.addEventListener("dragstart",ev=>{
      DRAG=el.dataset.edita; el.classList.add("dragging");
      try{ ev.dataTransfer.setData("text/plain",DRAG); ev.dataTransfer.effectAllowed="move"; }catch(e){}
    });
    el.addEventListener("dragend",()=>{ el.classList.remove("dragging"); DRAG=null;

```

```

    root.querySelectorAll(".ecell").forEach(c=>c.classList.remove("drop","nodrop")); });
});
root.querySelectorAll(".ecell").forEach((cell,i)=>{
  cell.addEventListener("dragover",ev=>{
    if(!DRAG) return; ev.preventDefault();
    const t=cellTarget(cell); if(!t) return;
    const a=getWeek().assignments.find(x=>x.id===DRAG); if(!a) return;
    const e=empById(t.empId);
    const sh=shiftById(a.shiftId);
    const [aM,bM]=shiftMinutes(sh);
    const clean = e && availOk(e,t.dow,aM,bM) && !onTimeOff(e.id,t.date)
      && !getWeek().assignments.some(x=>x.id!==a.id && x.empId===t.empId && x.date===t.date);
    cell.classList.toggle("drop",clean); cell.classList.toggle("nodrop",!clean);
  });
  cell.addEventListener("dragleave",()=>cell.classList.remove("drop","nodrop"));
  cell.addEventListener("drop",ev=>{
    ev.preventDefault(); cell.classList.remove("drop","nodrop");
    if(!DRAG) return;
    const t=cellTarget(cell); if(!t) return;
    const wk=getWeek(); const a=wk.assignments.find(x=>x.id===DRAG); if(!a) return;
    if(wk.assignments.some(x=>x.id!==a.id && x.empId===t.empId && x.date===t.date)){
      toast((empById(t.empId)||{name:"They"}).name+" is already on that day"); return;
    }
    a.empId=t.empId; a.date=t.date; a.dow=t.dow;
    const e=empById(t.empId);
    if(e && !e.roleIds.includes(a.roleId)) a.roleId = e.roleIds[0]||null;
    applyBreaks(a); a.locked=true;
    wk.assignments.sort((x,y)=>x.date.localeCompare(y.date)||x.startMin-y.startMin);
    save(); render();
    const V=validate().filter(v=>v.sev==="crit" && v.empId===t.empId && v.date===t.date);
    toast(V.length? "Moved - "+V[0].title.split(" - ")[0].toLowerCase() : "Moved and locked");
  });
});
}
function cellTarget(cell){
  const row=cell.parentElement;
  const cells=...row.children;
  const idx=cells.indexOf(cell)-1;
  const dates=weekDates();
  if(idx<0||idx>6) return null;
  const rows=...row.parentElement.querySelectorAll(".erow");
  const rowIdx=rows.indexOf(row);
  const roster=S.employees.filter(e=>e.active);
  const emp=roster[rowIdx];
  if(!emp) return null;
  return {empId:emp.id, date:dates[idx].date, dow:dates[idx].dow};
}

/* ----- go ----- */
boot();
</script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js"></script>

```

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf-autotable/3.8.2/jspdf.plugin.autotable.min.js"></script>
```